

Simulating phenotypic data

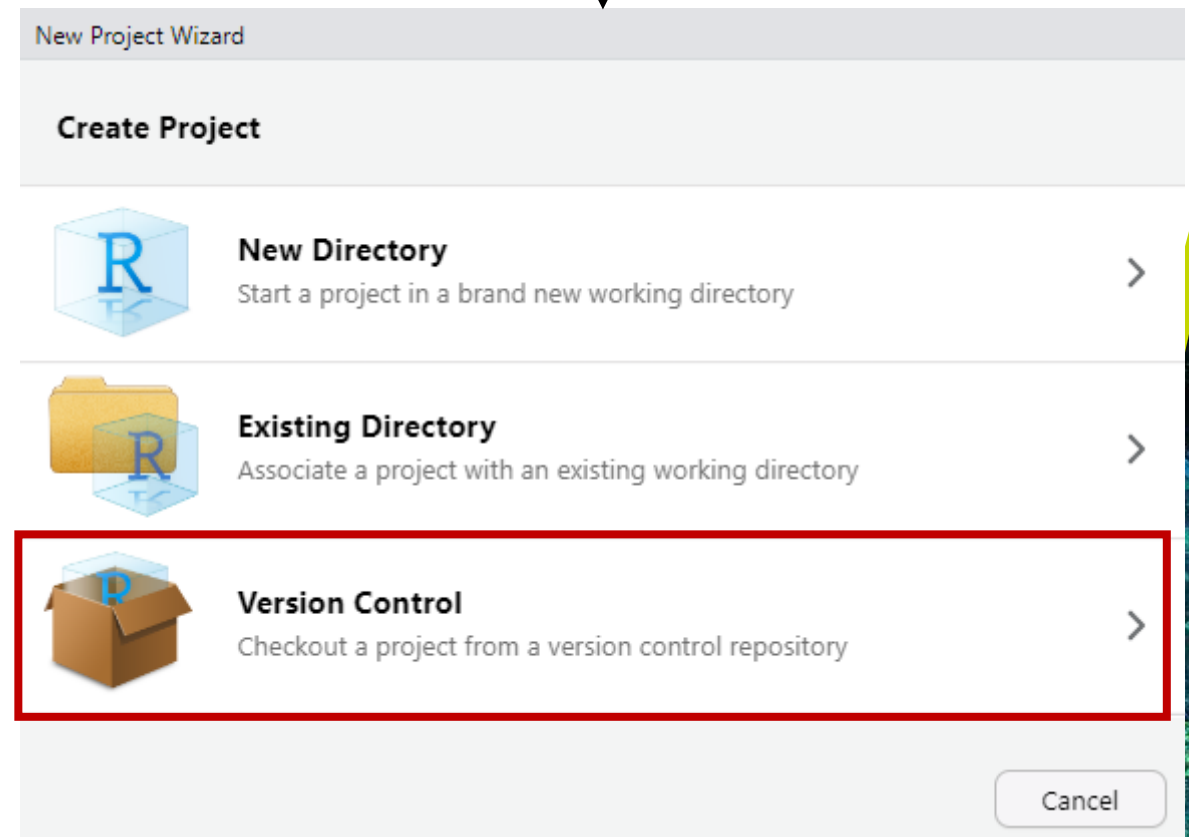
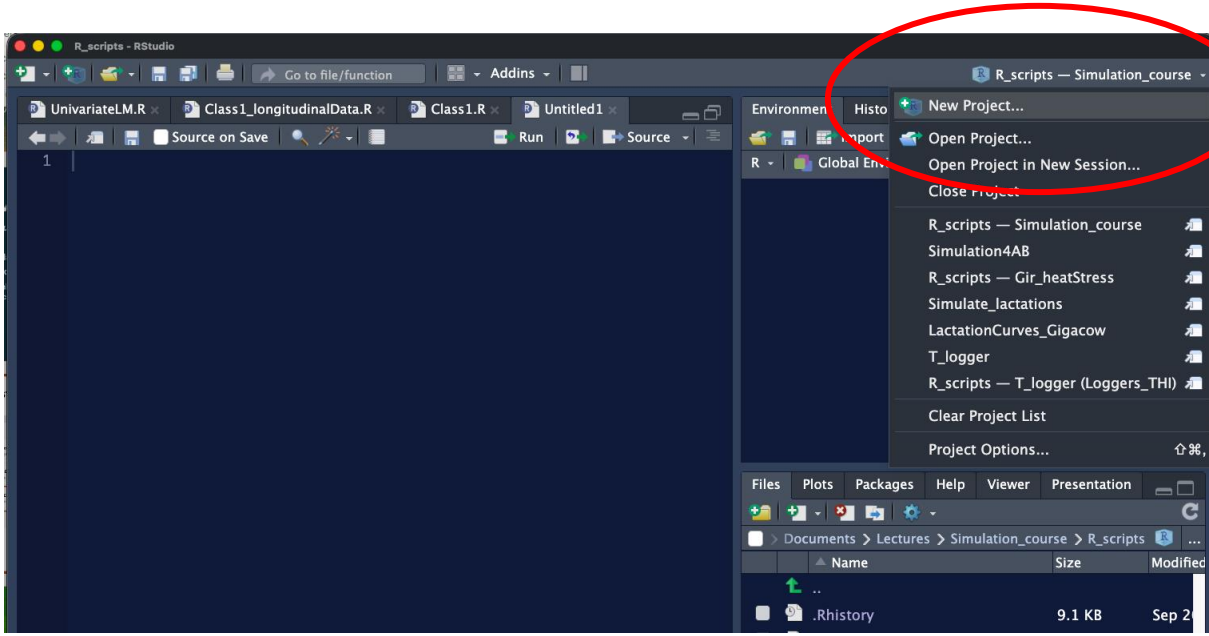
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Overview

This lecture will be in 3 parts:

1. Simulating cross-sectional phenotypes
2. Including systematics effects in the phenotype
3. Modeling longitudinal (time-series) data
 - a) *Including the effect of the farm*

First of all, clone the Github repository



New Project Wizard

Back

Create Project from Version Control



Git

Clone a project from a Git repository



Subversion

Checkout a project from a Subversion repository



Cancel

New Project Wizard

Back

Clone Git Repository



Repository URL:

<https://github.com/Hector-Marina/Simulation4AB>

Project directory name:

Simulation4AB

Create project as subdirectory of:

Browse...

☒ Open in new session

Create Project

Cancel

Environment

History

Connections

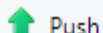
Git

Tutorial

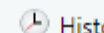


Commit

Pull



Push



History



More



main



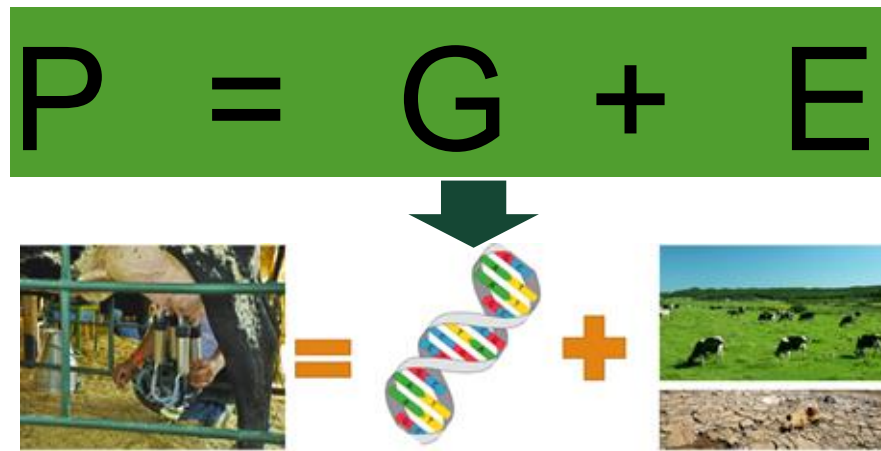
1. Cross-sectional data

Phenotype

What does it mean to simulate a phenotype?

A **phenotype** is an observable or measurable trait of an individual

Quantitative traits	Qualitative traits
Continue or discrete	Binary, nominal or ordinal



Why focus on quantitative traits?

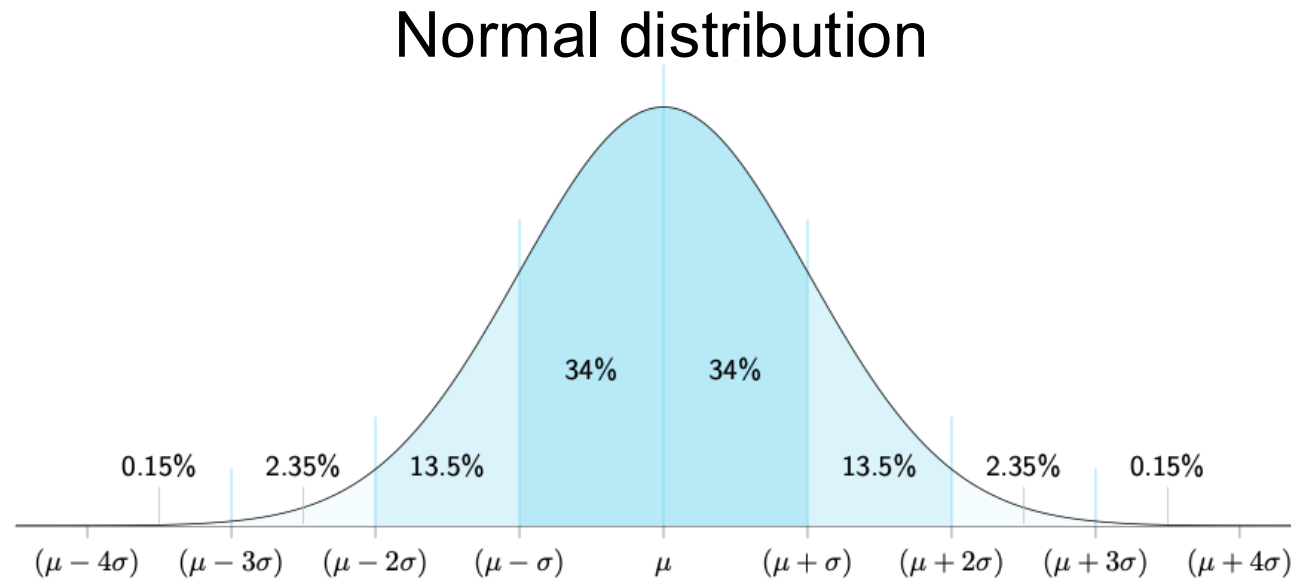
Quantitative traits are particularly important in animal breeding because many economically relevant production traits are measured numerically



These traits usually show **continuous variation among animals.**

From quantitative traits to a distribution

For many quantitative traits, the distribution can be **approximately bell-shaped**



$$p \sim N(\mu_p, \sigma_p^2)$$

p = phenotype

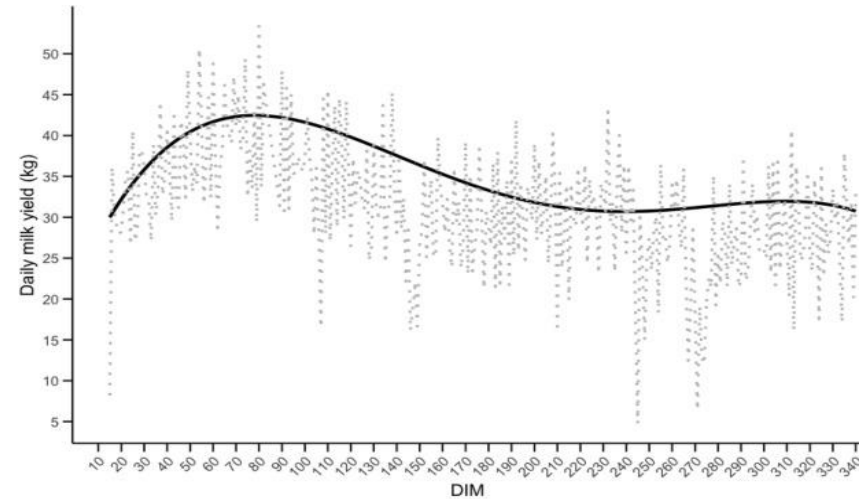
Data Structure: Cross-sectional vs. Longitudinal data

Cross-sectional data



Animal	Observation
1	1
2	1
3	1

Longitudinal data



Animal	Observation
1	1
1	2
1	3

Simulating cross-sectional phenotypes



2. Simulating systematic effects

Systematic effects?



=



+



- Lactation number
- Year/season of calving
- Age
- Farm

$$p_{ijk} = \mu + \underbrace{lactation_i + calv_season_j + herd_k}_{\text{Systematic effects}} + e_l$$

Systematic effects?



- Lactation number
- Year/season of calving
- Age
- Farm

$$y_{ijklm} = \mu + \underbrace{lactation_j + calv_season_k + herd_l}_{\text{Fix effects}} + \underbrace{e_m}_{\text{Random effect}}$$

Parameter		Contrast
Lactation number	1	0 (reference)
	2	8.9
	3	10.4
	4	10.7
	5	9.2
	6	8.3
Calving season	winter	0
	spring	1.2
	summer	-0.8
	fall	-0.3
Herd	Values ranged from 1.5 to 11.7	

Data of 1205 lactation from 607 Swedish Dairy cows (SLU Gigacow*)

* <https://www.slu.se/en/research/research-infrastructure/research-infrastructure-at-slu-open-to-external-users/gigacow/>

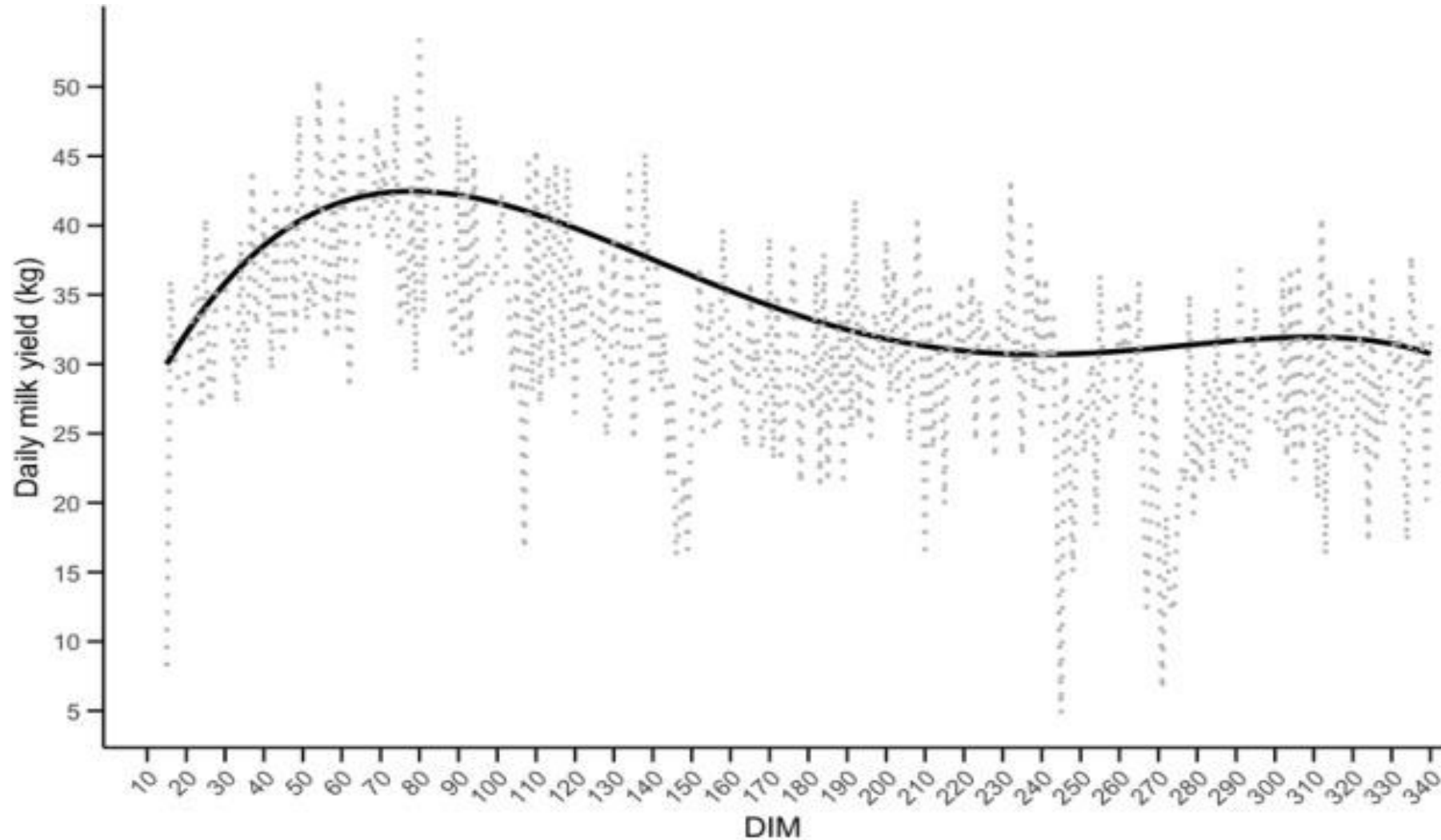
Including systematic effects



3. Time series data – longitudinal data

Time series data – longitudinal data

Multiple observations per animal over time

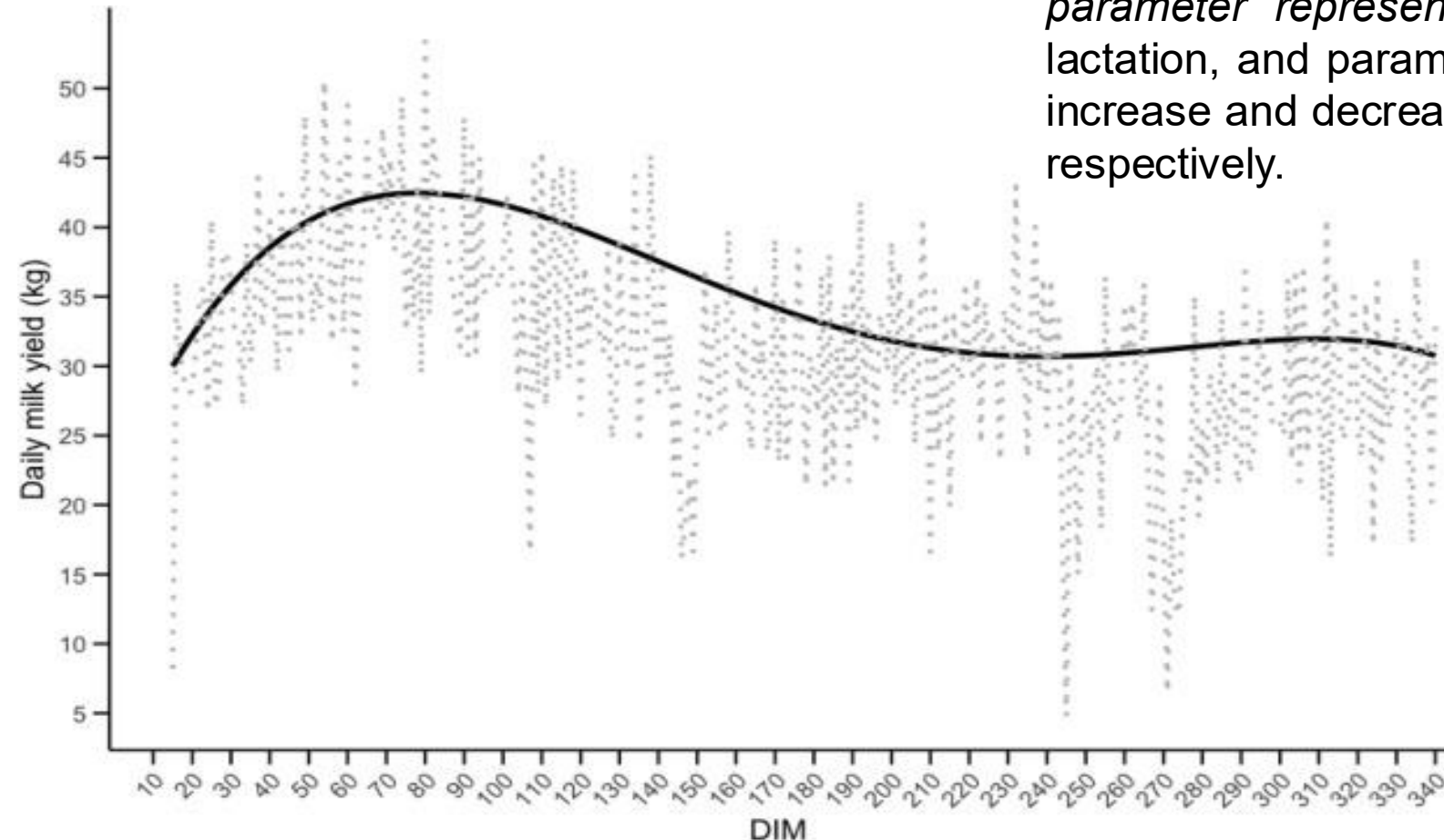


Lactation curve model

The Wood model

$$Y_{DIM} = a \times DIM^b \times \exp^{-c \times DIM}$$

where Y_{DIM} is the test-day record at a given DIM, a is the parameter representing production at the beginning of lactation, and parameters b and c are associated with the increase and decrease of the slopes of the lactation curve, respectively.

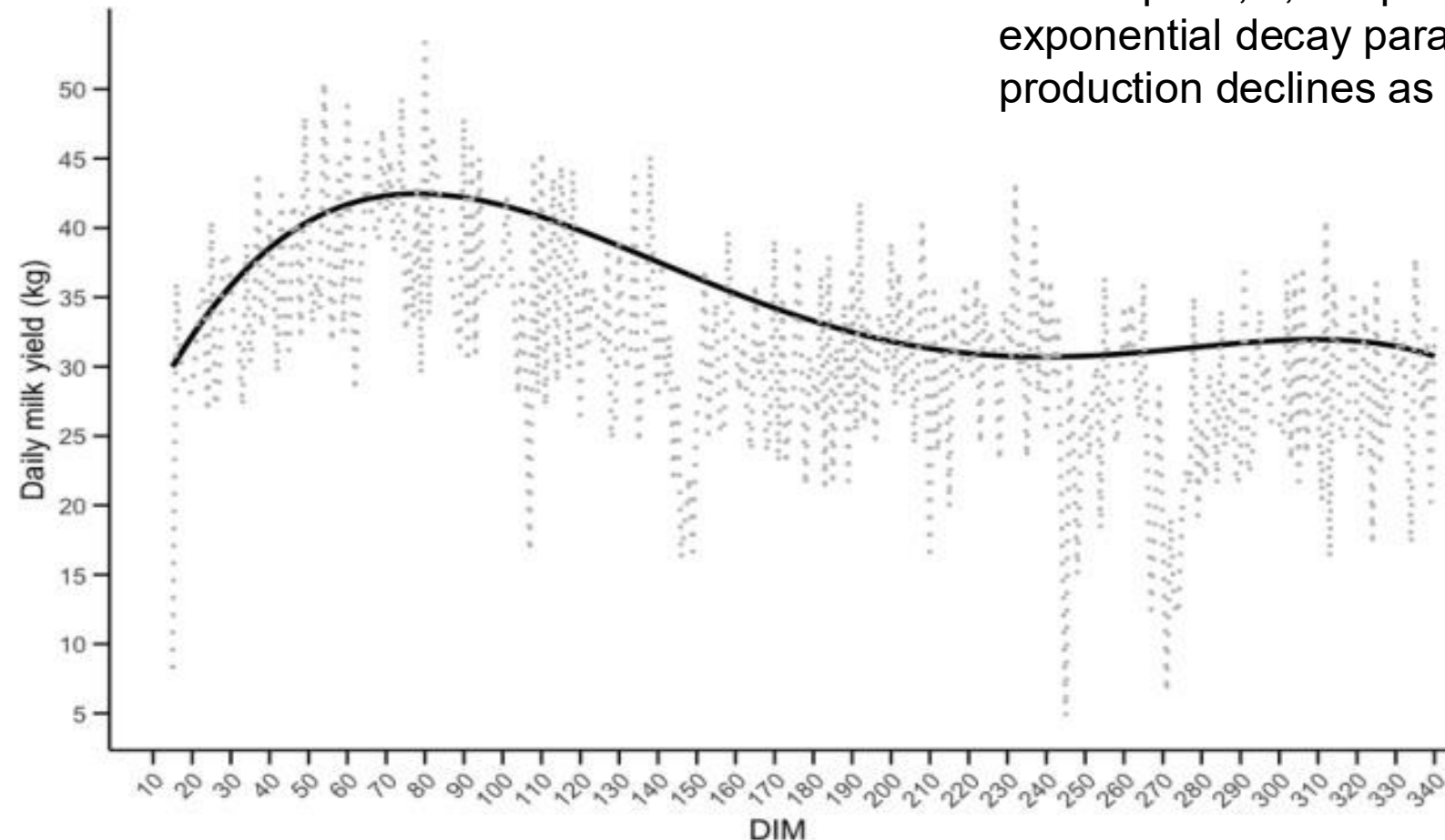


Wilmink equation

$$a + b * \exp(-k * t) + c * t$$

$$Y_{DIM} = a + b * \exp^{-k * DIM} + c * DIM$$

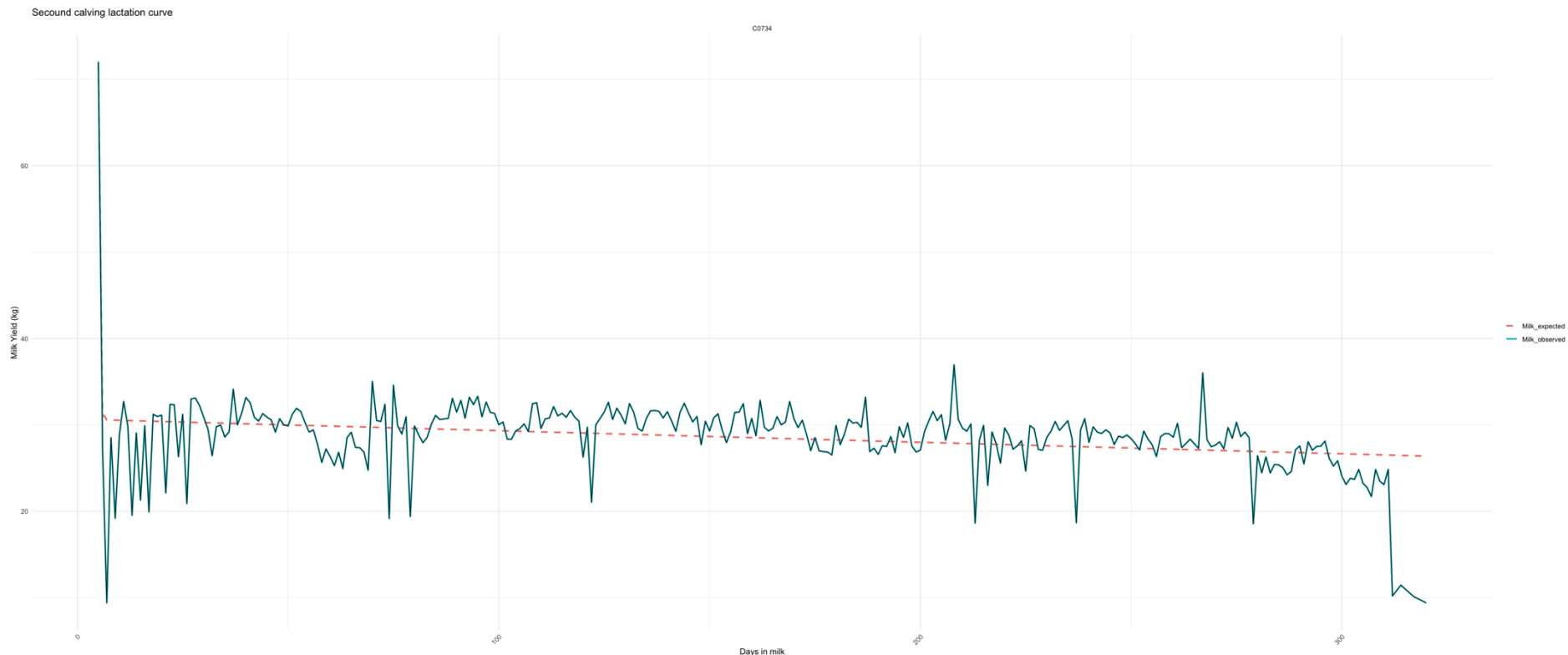
a , represent the level of production; b , the production increase toward peak; c , the production decrease after the peak. The exponential decay parameter k , which determines how fast production declines as time goes by.



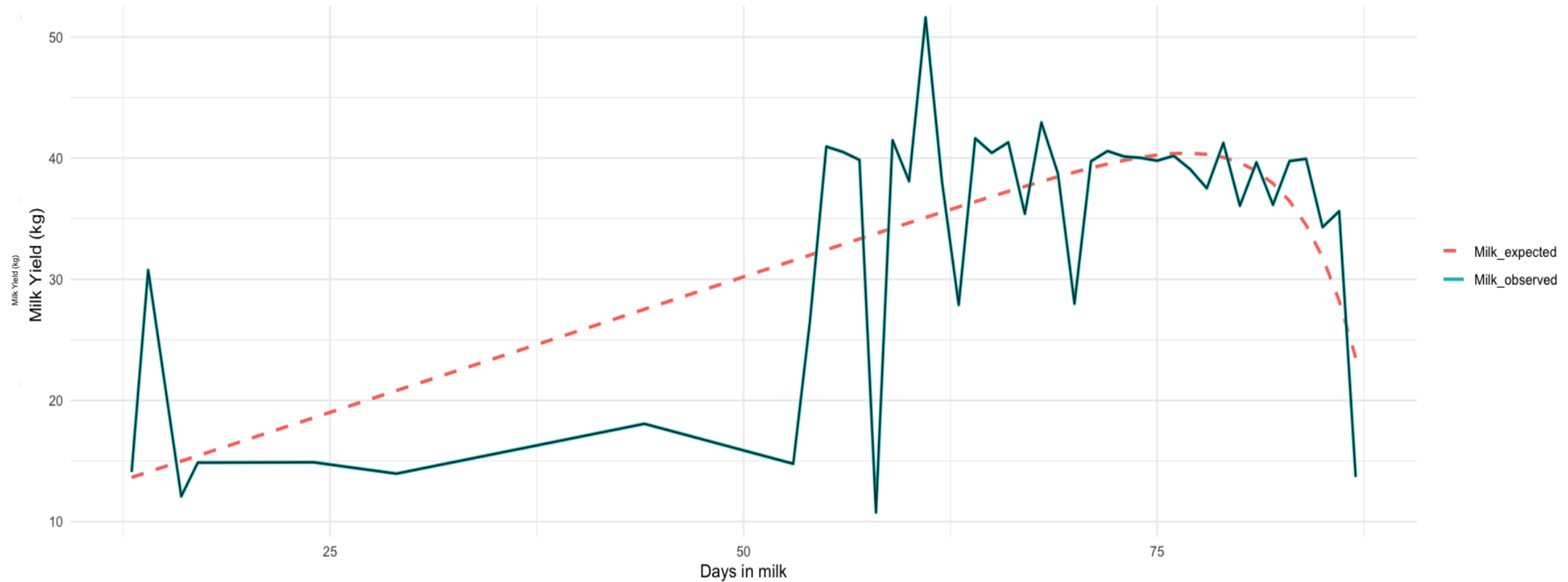
- We faced different problems to find the most ideal parameters to simulate data..

- Example: Atypical lactations cruves

Performing a graphical inspection of curves with extreme estimated parameters for Wilmink



- Extreme values for the Wilmink parameters



For simulation lactations curves

Wilmink lactations curves parameters

Parameter	Min	Max	After removing outliers	
			Min	Max
a	-62.01	852.40	-107.68	768.19
b	-3.235e+11	9.800e+01	-740.66	143.45
c	-0.93	0.86	-0.87	0.86
k	-0.11	1.33	-0.11	0.58

Simulating daily milk yield using Wood model





- Activity: Include the systematic effect of the farm
Create 5 farms using the following contrast for them

Farm 1=0

Farm 2 = -2

Farm 3= 1

Farm 4 = 2

Farm 5 = -1



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